



M-Tech Production & Marketing

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Experiment 5

Lab Tasks

Internship Program — Batch 2026

Submitted By:	Abdullah Rafique (MTech-AI260120)
Department:	AI / ML
Submitted To:	Ceo Moazzan
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1. Experiment 5 – Lab Tasks

File: *Experiment_5.py*

2. Viva Questions

1. What is the difference between Regression and Classification?

Regression and Classification are two major types of supervised Machine Learning algorithms.

Regression is used when the target output is a **continuous numerical value**, such as predicting house prices, temperatures, or examination scores. The model produces a numeric value as its prediction.

Classification, on the other hand, is used when the target output is a **discrete category or class**, such as determining whether an email is spam or not spam, identifying the species of a flower, or detecting fraudulent transactions. The model predicts a class label instead of a numerical value.

Regression	Classification
Predicts continuous numerical values	Predicts discrete categories or classes
Output is a number	Output is a label
Example: House Price Prediction	Example: Spam Email Detection
Example: Temperature Prediction	Example: Disease Classification

2. What are R² Score and RMSE?

R² Score (Coefficient of Determination) measures how well a regression model explains the variation in the target variable. Its value ranges from **0 to 1**, where values closer to **1** indicate a better-fitting model.

RMSE (Root Mean Squared Error) measures the average prediction error between the actual and predicted values. It is expressed in the same units as the target variable.

- **Higher R² Score** → Better model performance.
- **Lower RMSE** → More accurate predictions.

Together, these metrics help evaluate the effectiveness of regression models.

3. How does the K-Nearest Neighbors (KNN) algorithm make predictions?

The **K-Nearest Neighbors (KNN)** algorithm predicts the output for a new data point by finding the **K closest data points** in the training dataset based on a distance metric, typically **Euclidean Distance**.

- For **Classification**, the algorithm assigns the class that appears most frequently among the nearest neighbors.
- For **Regression**, it predicts the average value of the nearest neighbors.

The value of **K** plays an important role in determining the model's accuracy and performance.

4. Why should a Machine Learning model be tested on unseen data?

A Machine Learning model should always be evaluated using **unseen (test) data** to determine its ability to generalize to new examples.

If the model is tested using the same data on which it was trained, it may simply memorize the training examples instead of learning meaningful patterns. This phenomenon is known as **overfitting**.

Testing on unseen data provides a realistic measure of how well the model will perform in real-world applications and helps identify whether the model has learned the underlying relationships within the data.

5. What is Accuracy, and why can it sometimes be misleading?

Accuracy is the proportion of correct predictions made by a model compared to the total number of predictions.

Formula:

$$\text{Accuracy} = (\text{Correct Predictions} \div \text{Total Predictions}) \times 100$$

Although accuracy is easy to understand, it may produce misleading results when working with **imbalanced datasets**.

For example, if **95%** of emails are not spam, a model that always predicts "**Not Spam**" will achieve **95% accuracy**, even though it completely fails to detect spam emails.

For such datasets, evaluation metrics such as **Precision**, **Recall**, and **F1-Score** provide a more reliable assessment of model performance.

6. Which Machine Learning algorithms can perform both Regression and Classification?

Several Machine Learning algorithms can solve both regression and classification problems. Two of the most commonly used algorithms are:

Decision Tree

- Performs **Classification** by predicting the majority class within each leaf node.
- Performs **Regression** by predicting the average target value within each leaf node.

K-Nearest Neighbors (KNN)

- Performs **Classification** using majority voting among the nearest neighbors.
- Performs **Regression** by averaging the target values of the nearest neighbors.

These algorithms are flexible and can be adapted to different prediction tasks depending on the problem being solved.

3. Multiple Choice Questions (MCQs)

Q1. Predicting whether an email is spam or not spam is an example of:

Answer: (B) Classification

Q2. A perfect R^2 Score is:

Answer: (C) 1

Q3. The K-Nearest Neighbors (KNN) algorithm makes predictions using:

Answer: (B) K nearest points

Q4. A lower RMSE value indicates:

Answer: (B) Better model performance

Q5. Which algorithm is commonly used for Classification?

Answer: (B) K-Nearest Neighbors (KNN)

4. Intern Information

Field	Details
Intern Name	Abdullah Rafique
Supervisor Name	Mr. Mohazin Zahid
Date	08 July, 2026



Field	Details
Domain	AI/ML Intern
Intern Signature	_____A.B.D_____